COS30045 Data Visualisation

Assessments

Agenda

- Data Visualisation Critique
- Data Visualisation Project

Visualisation Critique

Aim: Critique visualisation according to data visualisation guidelines

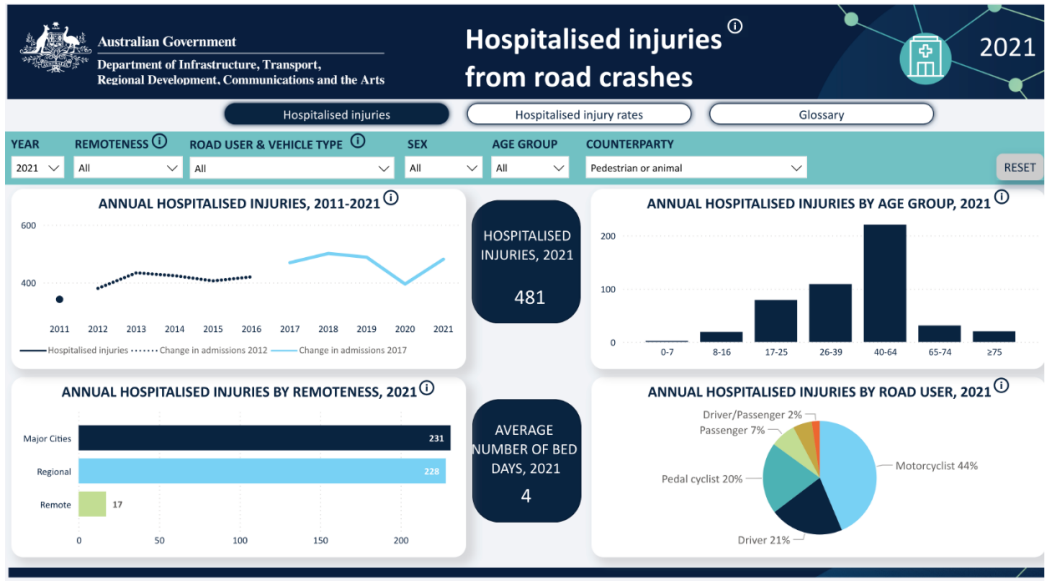
Due: Week 5 (just before Easter)

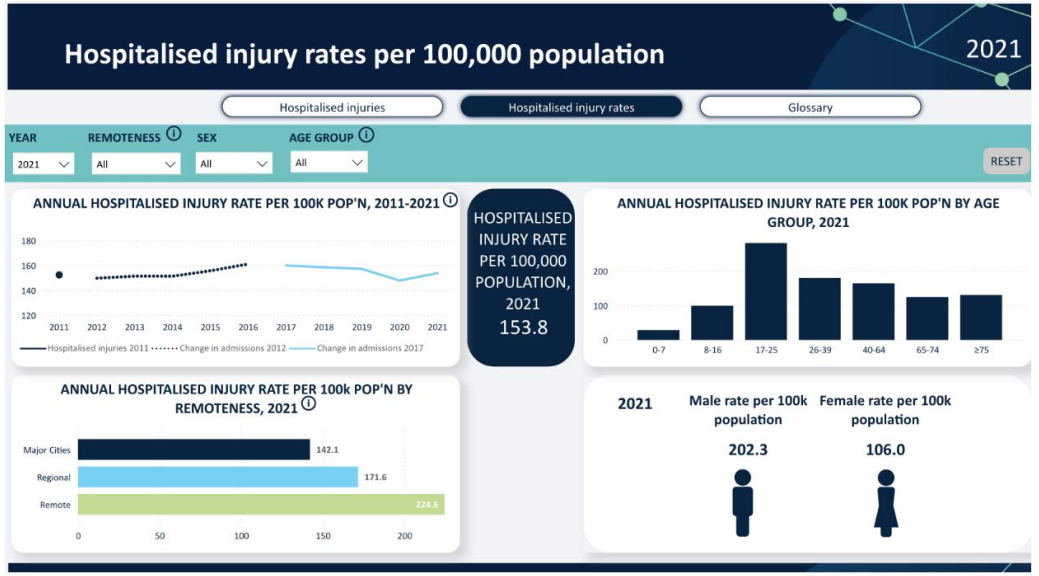
Structure

- Introduction
- Data Processing and Governance
- Visualisation Design
- Recommendations

Must be on specified visualisation

(see Canvas for this semester's visualisation)

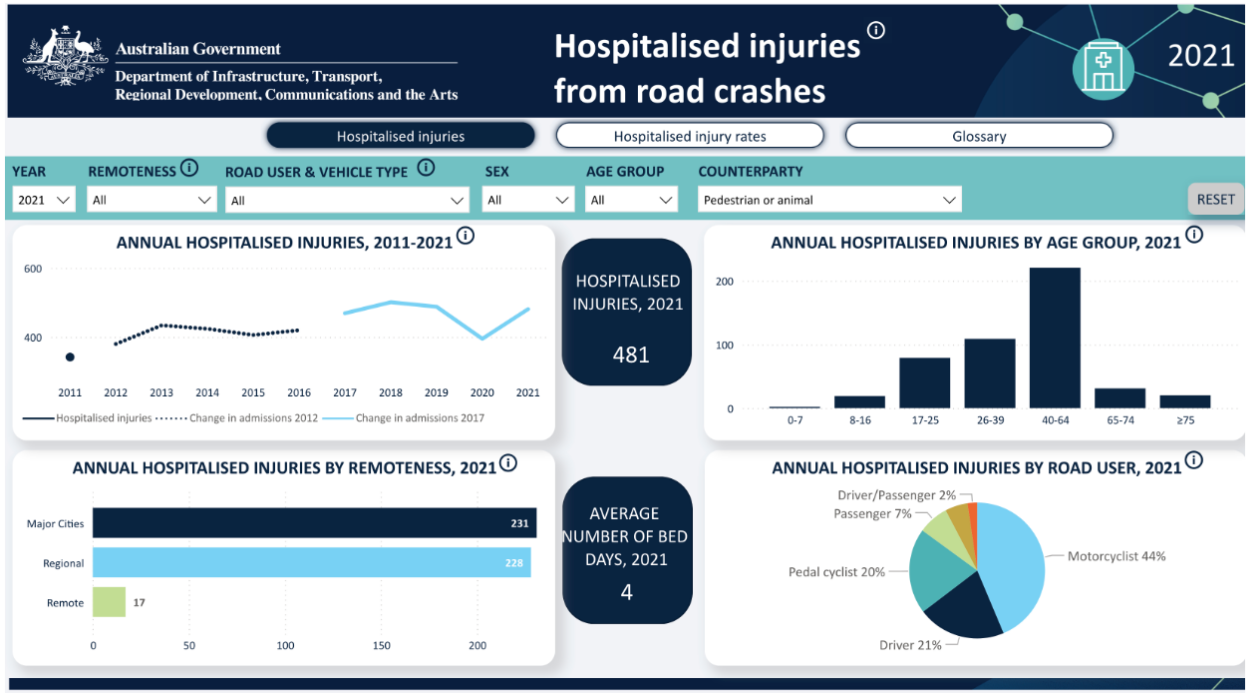




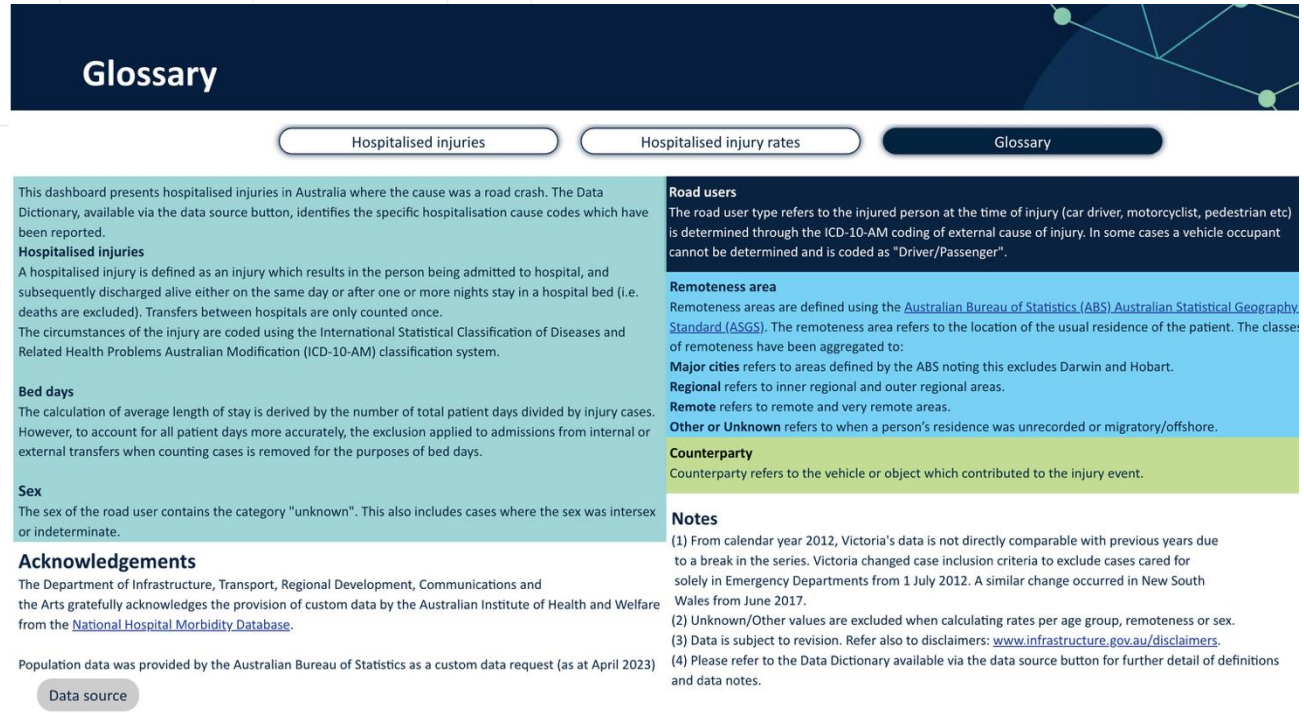
Site

Assignment

Criteria	Ratings					Pts
Identification of Audience and their needs	**5 to >3.5 Pts Excellent** Demonstrates excellent understanding of context of use. Clearly defines the intended audience, considering user expertise, and specific examples are given about application of the data to problems. Cites academic or policy documents to support analysis.	**3.5 to >3.0 Pts Good** Identifies the target audience in general terms. Includes some with discussion of their specific needs based on analysis of context of use.	**3 to >2.5 Pts Satisfactory** Mentions the audience and their needs, but lacks depth in analysis based on context of use.	**2.5 to >2.25 Pts Limited** Briefly acknowledges the audience but description is superficial.	**2.25 to >0 Pts Unsatisfactory** No or very limited discussion of the intended audience or their needs.	5 pts



Data Processing	**5 to >3.5 Pts Excellent** Demonstrates an excellent understanding of the nature of the data, its source, limitations and interpretation. Provides a thorough examination of the data dictionary and any cleaning or transformations applied and their impact on visualisation.	**3.5 to >3.0 Pts Good** Demonstrates good understanding of the data dictionary and process of data collection and transformation. Some attempt is made to more deeply investigate the nature of the data, limitations in its collection and it's impact on the visualisation.	**3 to >2.5 Pts Satisfactory** Demonstrates some understanding of the data dictionary and the transformations applied to the results or limitations in data collection method.	**2.5 to >2.25 Pts Limited** Superficially discusses the contents of the data dictionary without any commentary or interpretation.	**2.25 to >0 Pts Unsatisfactory** Appears not to have read or understood data dictionary.	5 pts



Glossary

Hospitalised injuries | Hospitalised injury rates | Glossary

This dashboard presents hospitalised injuries in Australia where the cause was a road crash. The Data Dictionary, available via the data source button, identifies the specific hospitalisation cause codes which have been reported.

Hospitalised injuries
A hospitalised injury is defined as an injury which results in the person being admitted to hospital, and subsequently discharged alive either on the same day or after one or more nights stay in a hospital bed (i.e. deaths are excluded). Transfers between hospitals are only counted once.
The circumstances of the injury are coded using the International Statistical Classification of Diseases and Related Health Problems Australian Modification (ICD-10-AM) classification system.

Bed days
The calculation of average length of stay is derived by the number of total patient days divided by injury cases. However, to account for all patient days more accurately, the exclusion applied to admissions from internal or external transfers when counting cases is removed for the purposes of bed days.

Sex
The sex of the road user contains the category "unknown". This also includes cases where the sex was intersex or indeterminate.

Road users
The road user type refers to the injured person at the time of injury (car driver, motorcyclist, pedestrian etc) is determined through the ICD-10-AM coding of external cause of injury. In some cases a vehicle occupant cannot be determined and is coded as "Driver/Passenger".

Remoteness area
Remoteness areas are defined using the Australian Bureau of Statistics (ABS) Australian Statistical Geography Standard (ASGS). The remoteness area refers to the location of the usual residence of the patient. The classes of remoteness have been aggregated to:
Major cities refers to areas defined by the ABS noting this excludes Darwin and Hobart.
Regional refers to inner regional and outer regional areas.
Remote refers to remote and very remote areas.
Other or Unknown refers to when a person's residence was unrecorded or migratory/offshore.

Counterparty
Counterparty refers to the vehicle or object which contributed to the injury event.

Acknowledgements

The Department of Infrastructure, Transport, Regional Development, Communications and the Arts gratefully acknowledges the provision of custom data by the Australian Institute of Health and Welfare from the National Hospital Morbidity Database.

Population data was provided by the Australian Bureau of Statistics as a custom data request (as at April 2023)

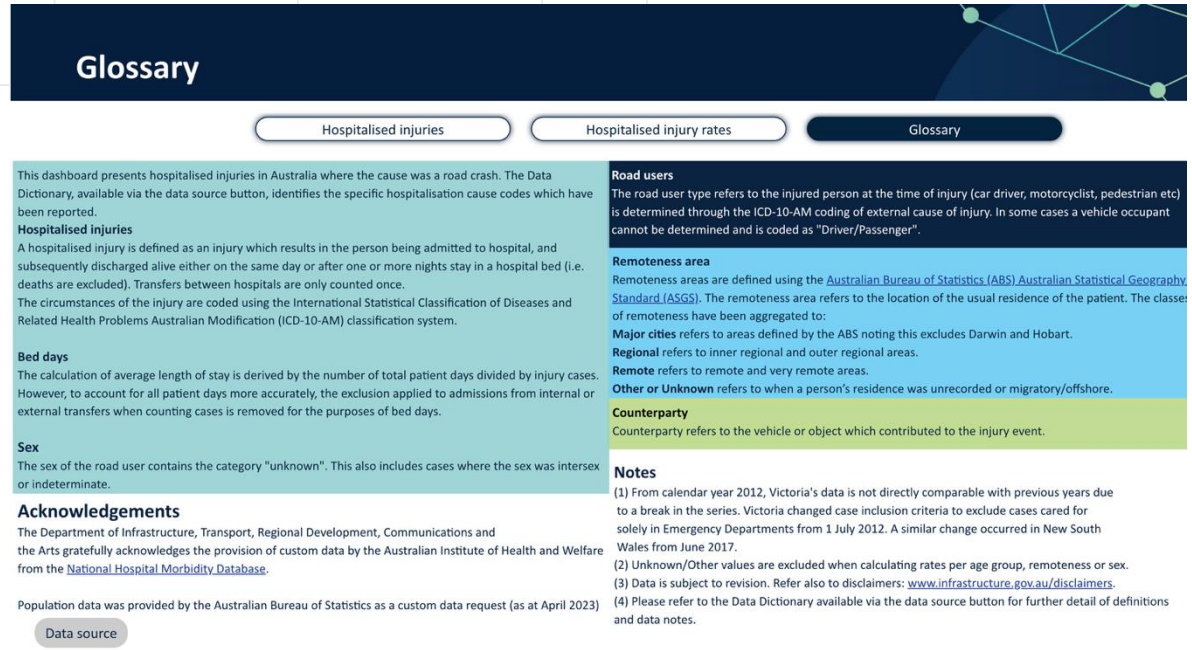
Data source

Notes

(1) From calendar year 2012, Victoria's data is not directly comparable with previous years due to a break in the series. Victoria changed case inclusion criteria to exclude cases cared for solely in Emergency Departments from 1 July 2012. A similar change occurred in New South Wales from June 2017.
(2) Unknown/Other values are excluded when calculating rates per age group, remoteness or sex.
(3) Data is subject to revision. Refer also to disclaimers: www.infrastructure.gov.au/disclaimers.
(4) Please refer to the Data Dictionary available via the data source button for further detail of definitions and data notes.

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Data Governance, Bias & Ethical Considerations	**3 to >2.5 Pts Excellent** Critically evaluates evidence of data governance and potential biases in data selection, processing, or visualisation choices, considering ethical implications.	**2.5 to >2.0 Pts Good** Identifies potential biases and addresses data governance but does not fully explore their impact. Some use of supporting references.	**2 to >1.8 Pts Satisfactory** Mentions bias but lacks depth in discussion. Some description of data governance.	**1.8 to >1.35 Pts Limited** Acknowledges bias without meaningful critique. Very brief mention of data governance.	**1.35 to >0 Pts Unsatisfactory** No mention of data governance, bias or ethical concerns.	3 pts



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Visualisation Elements	**5 to >3.5 Pts Excellent** Explains how data is mapped to visual elements and evaluates if the encoding choices are appropriate and effective. Good use of supporting references.	**3.5 to >3.0 Pts Good** Discusses encoding choices with some analysis of their effectiveness.	**3 to >2.5 Pts Satisfactory** Mentions encoding but does not critique effectiveness.	**2.5 to >2.25 Pts Limited** Briefly touches on encoding without meaningful analysis. Demonstrates poor understanding of graphical encoding of data. No or limited use of annotated screenshots to illustrate discussion.	**2.25 to >0 Pts Unsatisfactory** No discussion of data representation or encoding.	5 pts
Use of Visual Elements (Colour, Labels, Chart Type, etc.)	**5 to >3.5 Pts Excellent** Critically examines how well colour, labels, chart type, and other elements enhance (or not) readability and understanding. Demonstrates excellent knowledge of visualisation design principles in analysis of visual elements.	**3.5 to >3.0 Pts Good** Identifies key visual elements and provides some discussion on their effectiveness and adherence to good visualisation design principles.	**3 to >2.5 Pts Satisfactory** Describes use of visual elements accurately and demonstrates some awareness of visualisation design guidelines.	**2.5 to >2.25 Pts Limited** Limited description of visual elements and poor application of visual design guidelines to discussion of visualisation effectiveness. No or limited use of annotated screenshots to illustrate discussion.	**2.25 to >0 Pts Unsatisfactory** No or very limited discussion of visual elements in relation to guidelines.	5 pts

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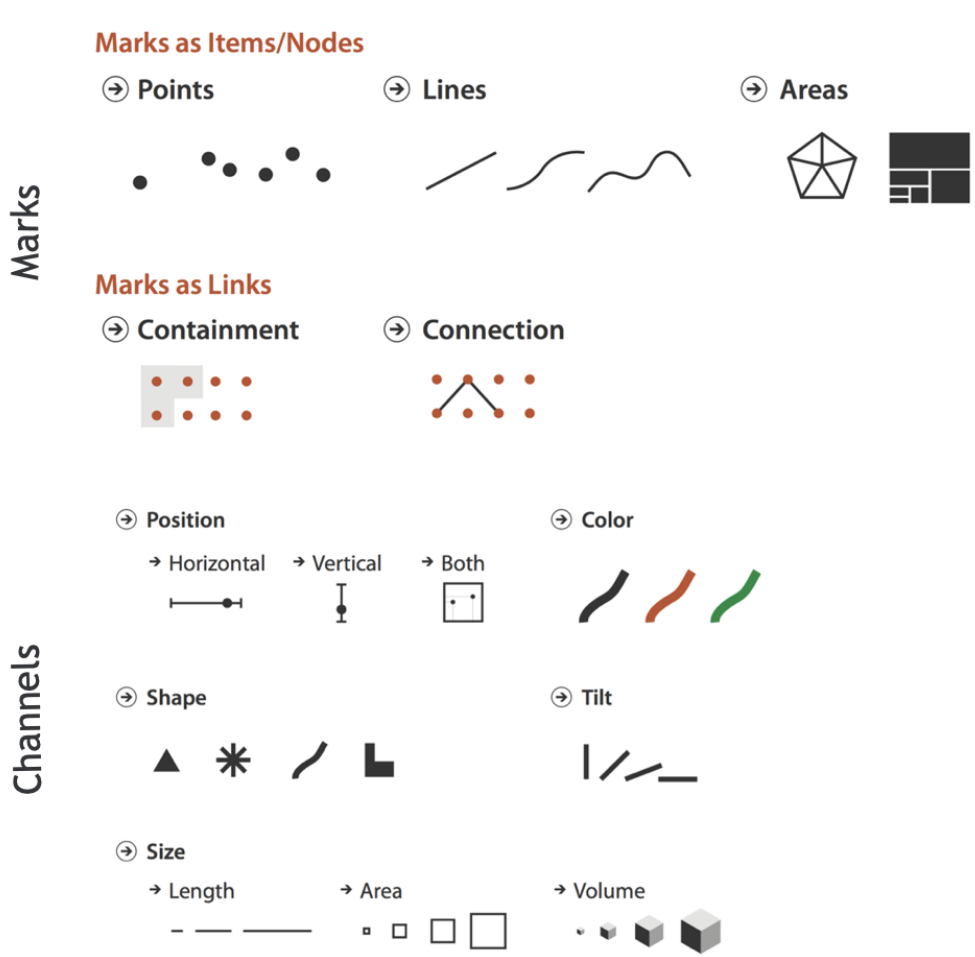
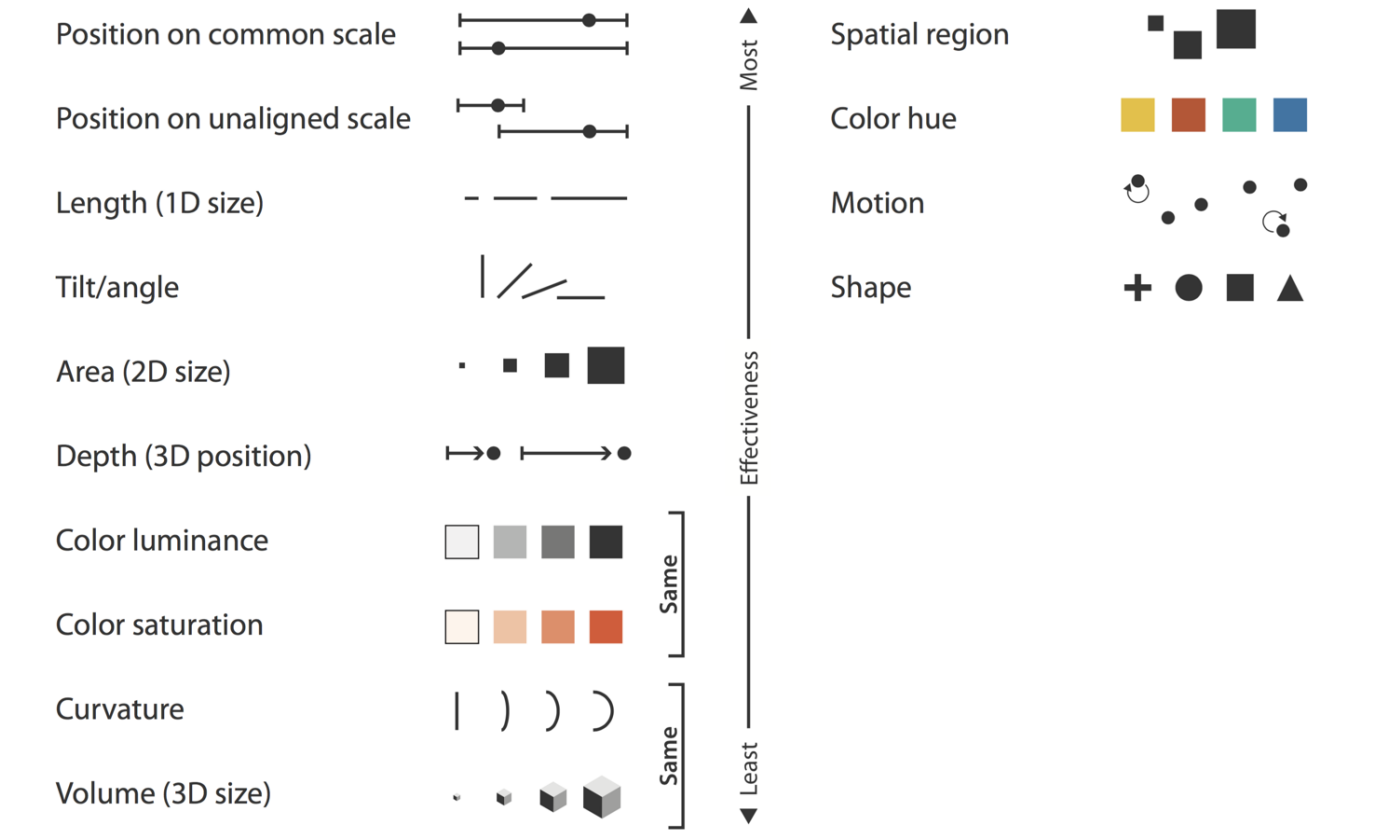


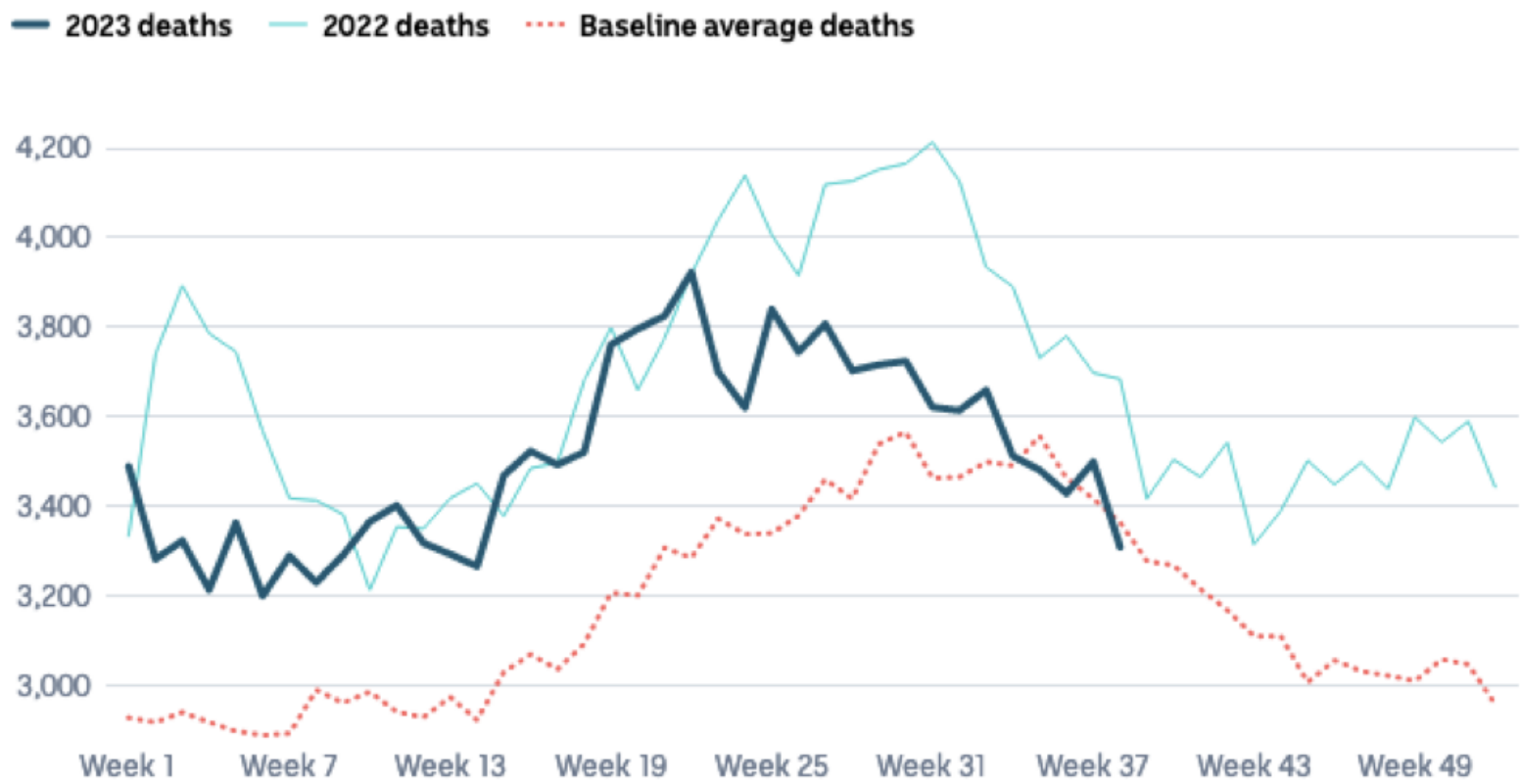
Figure 5.3. Visual channels control the appearance of marks.

Channels: Expressiveness Types and Effectiveness Ranks



How Australia's death rate compares to the baseline benchmark

Here's how the ABS says the number of deaths from January 3, 2022 to September 24, 2023 compares to the average baseline. The baseline is formulated from deaths between 2013 and 2019.

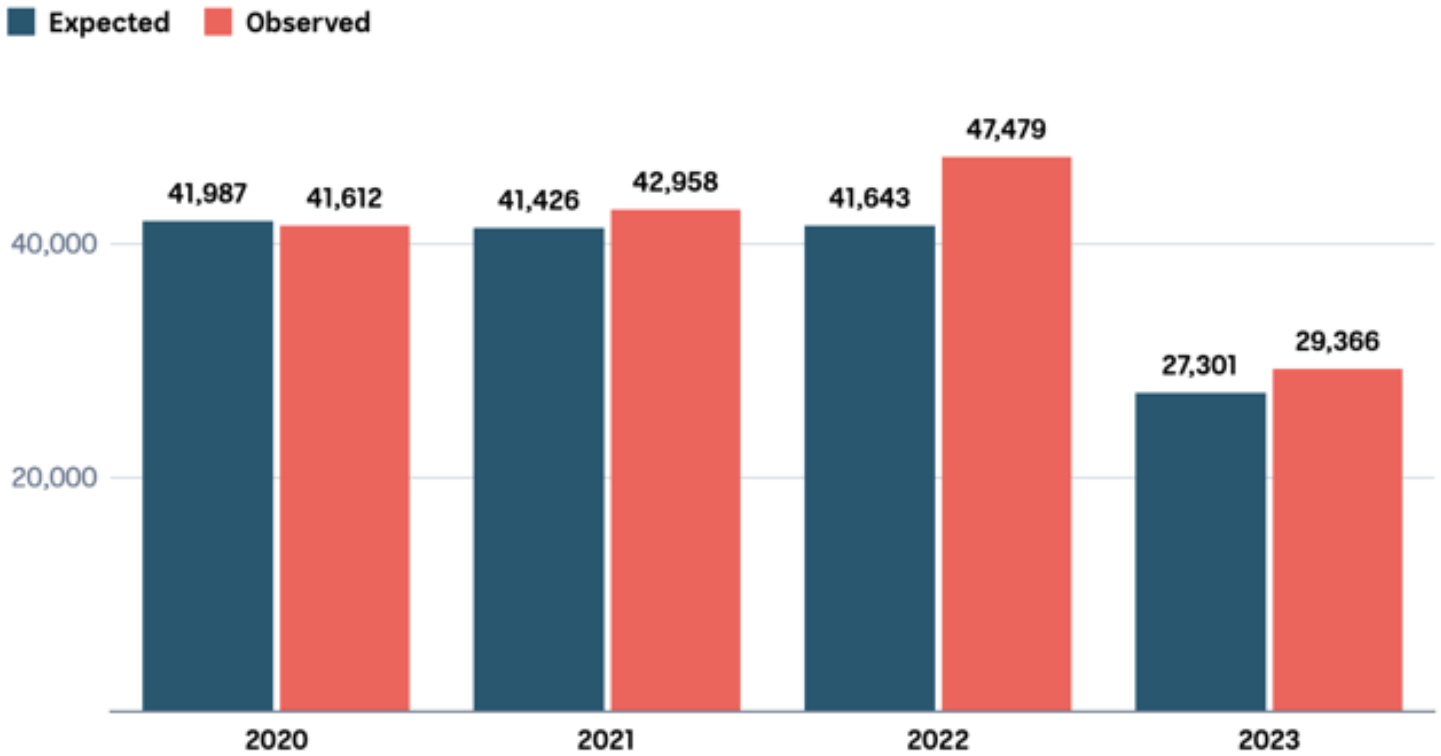


Source: ABS

https://www.abc.net.au/news/2023-12-20/mortality-rates-australia-covid-excess/103241640

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Excess mortality by year, Victoria

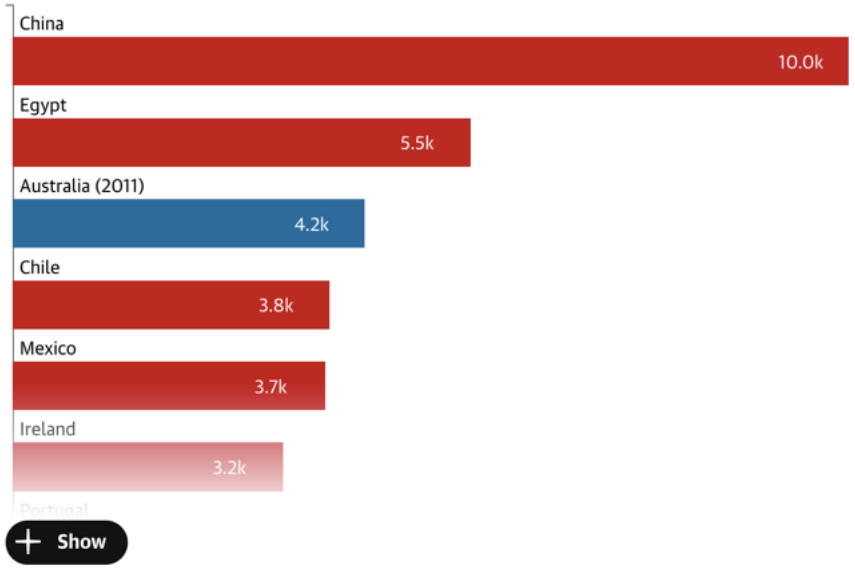


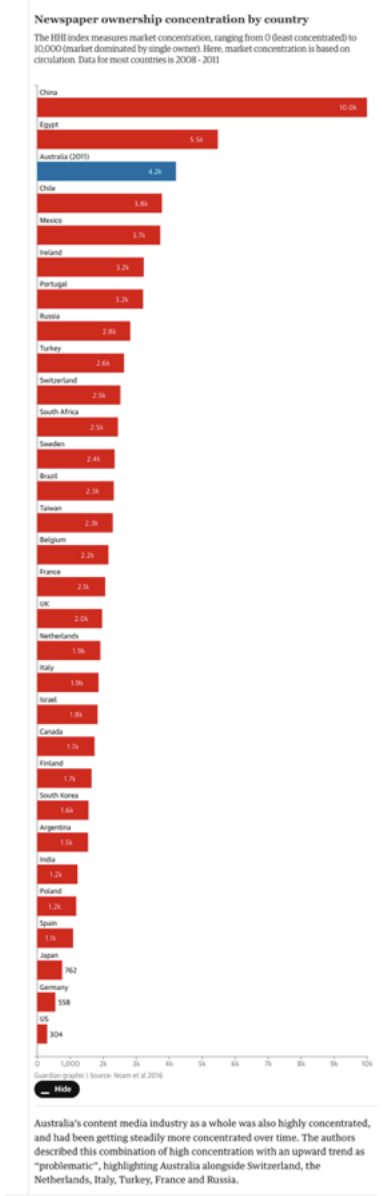
Source: ABS

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Newspaper ownership concentration by country

The HHI index measures market concentration, ranging from 0 (least concentrated) to 10,000 (market dominated by single owner). Here, market concentration is based on circulation. Data for most countries is 2008 - 2011

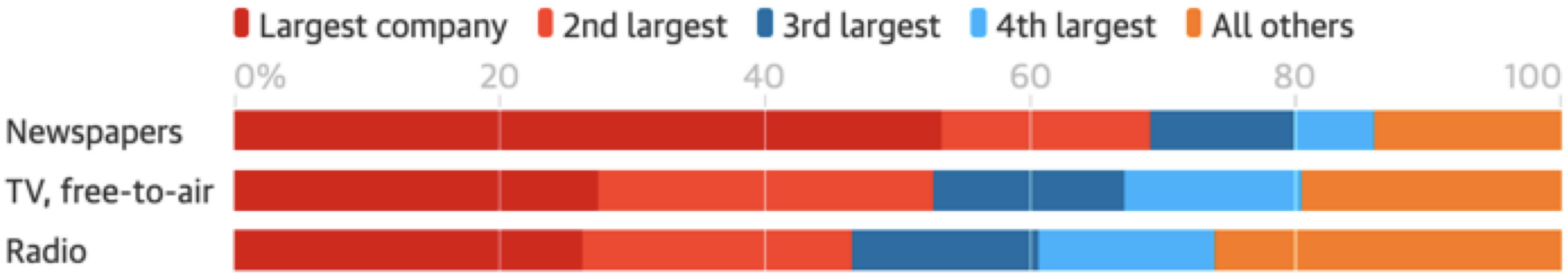




https://www.theguardian.com/news/datablog/2020/nov/13/australia-newspaper-ownership-is-among-the-most-concentrated-in-the-world

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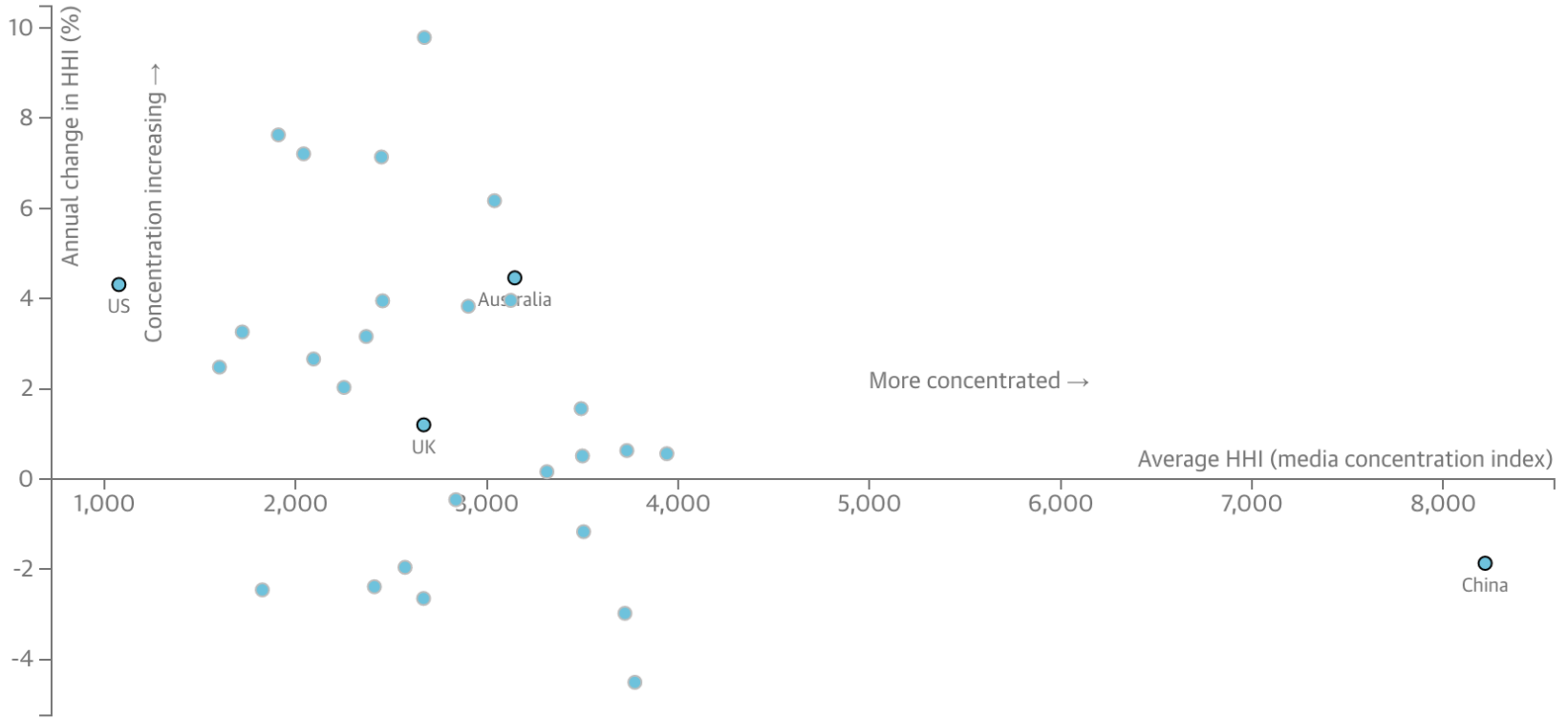
Per cent market share by revenue for top four companies in Australian media sectors



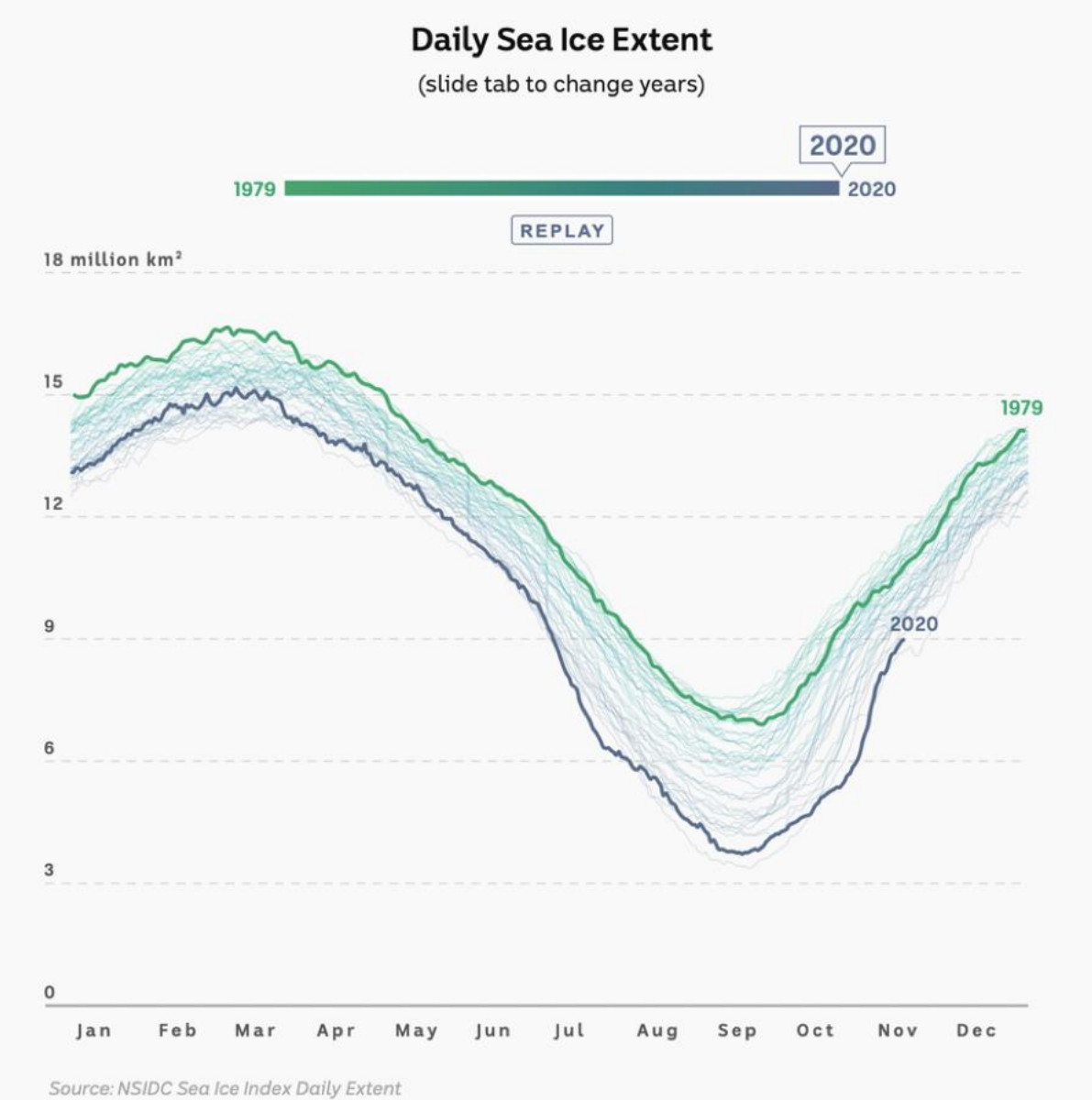
Guardian graphic | Data courtesy IBISWorld

https://www.theguardian.com/news/datablog/2020/nov/13/australia-newspaper-ownership-is-among-the-most-concentrated-in-the-world

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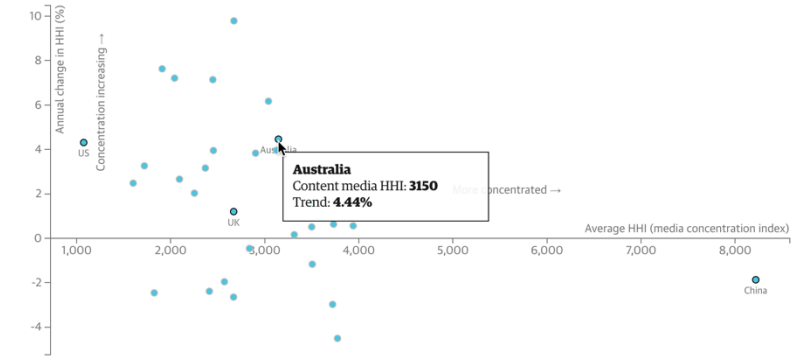
https://www.theguardian.com/news/datablog/2020/nov/13/australia-newspaper-ownership-is-among-the-most-concentrated-in-the-world



https://www.abc.net.au/news/2020-12-16/arctic-sea-ice-dramatic-transformation-as-seen-by-satellites/12961584?nw=0

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Implementation and interactivity	**5 to >4.0 Pts Excellent** Critically examines how well interactive elements function, assessing usability, responsiveness, efficiency, and technical limitations.	**4 to >3.5 Pts Good** Discusses usability and responsiveness of interactive elements with minor gaps in analysis.	**3.5 to >3.0 Pts Satisfactory** Mentions interactivity and assesses responsiveness but lacks detailed critique.	**3 to >2.25 Pts Limited** Briefly acknowledges interactivity without analysis. No or limited use of annotated screenshots to illustrate discussion.	**2.25 to >0 Pts Unsatisfactory** No or very limited discussion of interactivity, usability, performance or technical execution.	5 pts
Recommendations	**5 to >4.0 Pts Excellent** Recommendations directly address identified issues, are well-justified, feasible, and clearly improve clarity, usability, or accuracy. Demonstrates creativity and innovation.	**4 to >3.5 Pts Good** Recommendations align with critique, mostly feasible, and likely to improve the visualisation, but lack some depth or originality.	**3.5 to >3.0 Pts Satisfactory** Suggestions are somewhat relevant and practical but not fully developed or impactful.	**3 to >2.5 Pts Limited** Some recommendations are vague, impractical, or do not fully address critique findings.	**2.5 to >0 Pts Unsatisfactory** Recommendations are missing, unclear, impractical, or unrelated to the critique.	5 pts



Guardian graphic | Source: Who Owns the World's Media?

Data Visualisation Project

Aim: Design and build visualisations on the Police Enforcement website

Due: Week 13 (after semester)

Pair project (teams formed by tutor) in Week 8

Submission:

- Design Book
  - Purpose, data processing, design, evaluation…
- Website

See Canvas for details



One of the dashboards

Project Teams

Students will be required to **work on their own** if they:

- have NOT made an effort to be placed in a team before Week 9,
- have NOT meaningfully communicate with their team before Week 9 Class, or
- have NOT contributed to teamwork before Week 10 Stand-up.

Preference will be given to forming teams of students with similar levels of engagement and skill set.



Project Design Book

1 Introduction

- 1.1 Background and motivation
- 1.2 Visualisation purpose

2 Data

- 2.1 Data source and governance
- 2.2 Data processing and analysis
- 2.3 Data exploration

3 Visualisation design

- 3.1 Website design
- 3.2 Visualisation design
- 3.3 Interaction design

4 Iteration and validation

- 4.1 Testing and refinements
- 4.2 Usability evaluation

5 Conclusion and future improvements

References

Appendices

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Programming skills to practice for Project

Different arrangements of charts

Adding headings and paragraphs

Webpage layout design

Try changing colour of dots depending on TV type

Try adding some more lines to line chart (i.e., state prices)

Filters

Animations

Tooltips

